

University of Antwerp

Geographical Information Systems (GIS) course

Google Earth Engine Project

Istanbul Airport: Land Reclamation Analysis

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1 Introduction

Istanbul Airport, inaugurated in October 2018, is one of the world's largest aviation hubs with a capacity exceeding 200 million passengers annually [1]. Construction required extensive coastal modification on the Black Sea shoreline, including land reclamation from marine areas. Quantifying the extent of this land gain is essential for understanding the environmental and engineering scale of the project [2].

This analysis addresses the research question: *How much land had to be gained from the sea to build Istanbul Airport compared to 2012?* However, it is important to note that the term "gained from the sea" in this context is imprecise. The NDWI-based classification detects all water bodies within the study area, including both coastal marine waters and inland water features such as lakes, ponds, wetlands, and temporary water accumulations. Therefore, the analysis quantifies the total conversion of *water-covered areas to terrestrial land cover* between pre-construction (2012) and post-construction (2024) periods, encompassing both Black Sea coastal infilling and drainage of inland water bodies.

2 Data and Methods

2.1 Study Area

The study area encompasses a 6 km buffer around Istanbul Airport (41.26°N, 28.75°E) on the European Istanbul Black Sea coast.

2.2 Satellite Imagery Selection

Two Landsat datasets were selected to capture pre- and post-construction conditions (Figure 1):

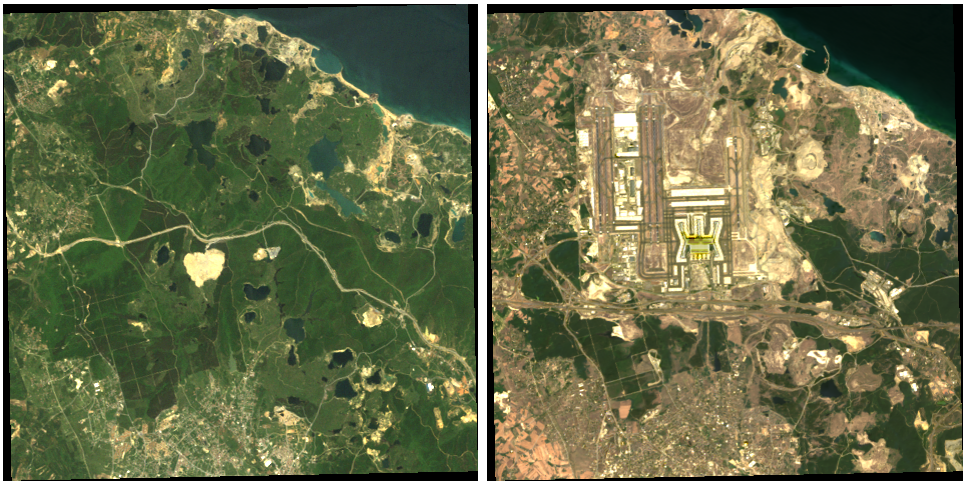


Figure 1: Satellite imagery. Left: Landsat 5 (2012). Right: Landsat 9 (2024).

- **Pre-construction (2012):** Landsat 5 TM (Collection 2, Level 2) imagery from 2011–2012, filtered for <50% cloud cover. Landsat 5 was chosen for its availability during this period and absence of scan line errors.
- **Post-construction (2024):** Landsat 9 OLI-2 (Collection 2, Level 2) imagery from 2023–2024, similarly filtered. Landsat 9 provides improved radiometric performance and consistency with Landsat 8/9 band designations.

2.3 NDWI Calculation and Water Classification

The Normalized Difference Water Index (NDWI) [3] [4] distinguishes water from land using the contrast between visible green and near-infrared (NIR) reflectance:

$$NDWI = \frac{Green - NIR}{Green + NIR} \tag{1}$$

Band assignments differed between sensors:

- Landsat 5: Green = Band 2, NIR = Band 4
- Landsat 9: Green = Band 3, NIR = Band 5

A threshold of NDWI > 0 was applied to classify water (1) versus land (0), producing binary rasters for 2012 and 2024.

2.4 Water-to-Land Conversion Quantification

Water-to-land conversion was identified by subtracting the 2024 water mask from the 2012 water mask: pixels classified as water in 2012 but land in 2024 represent converted areas. The total was computed using `reduceRegion` with `Reducer.sum()` at 30 m resolution (Landsat pixel size). Area was converted from pixels to km^2 by multiplying pixel count by 900 m^2 (30 m × 30 m) and dividing by 10^6.

3 Results

Figure 2 shows the spatial distribution of water-to-land conversion within the study area. Red pixels correspond to areas that were classified as water in 2012 and became terrestrial land by 2024. The pattern reveals changes concentrated around the airport footprint, including both Black Sea coastal infilling and drainage of inland water bodies (lakes, ponds, wetlands).

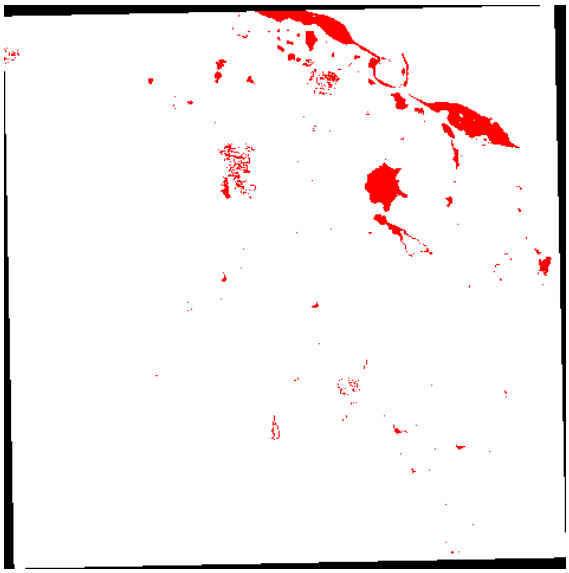


Figure 2: Water-to-land conversion (red) for Istanbul Airport construction. Analysis reveals approximately 2.4 km^2 of water-covered areas were converted to terrestrial land between 2012 and 2024, including both Black Sea coastal infilling and drainage of inland water bodies.

Quantitative results are summarized in Table 1.

Table 1: Water area and water-to-land conversion quantification.

Parameter	**2012**	**2024**
Water area (km^2)	8.73	6.32
Water-to-land conversion (km^2)	**2.41**	

The analysis reveals that approximately **2.41 km^2** of water-covered areas were converted to terrestrial land within this area between 2012 and 2024, reflecting drainage of inland water bodies and potential coastal modification.

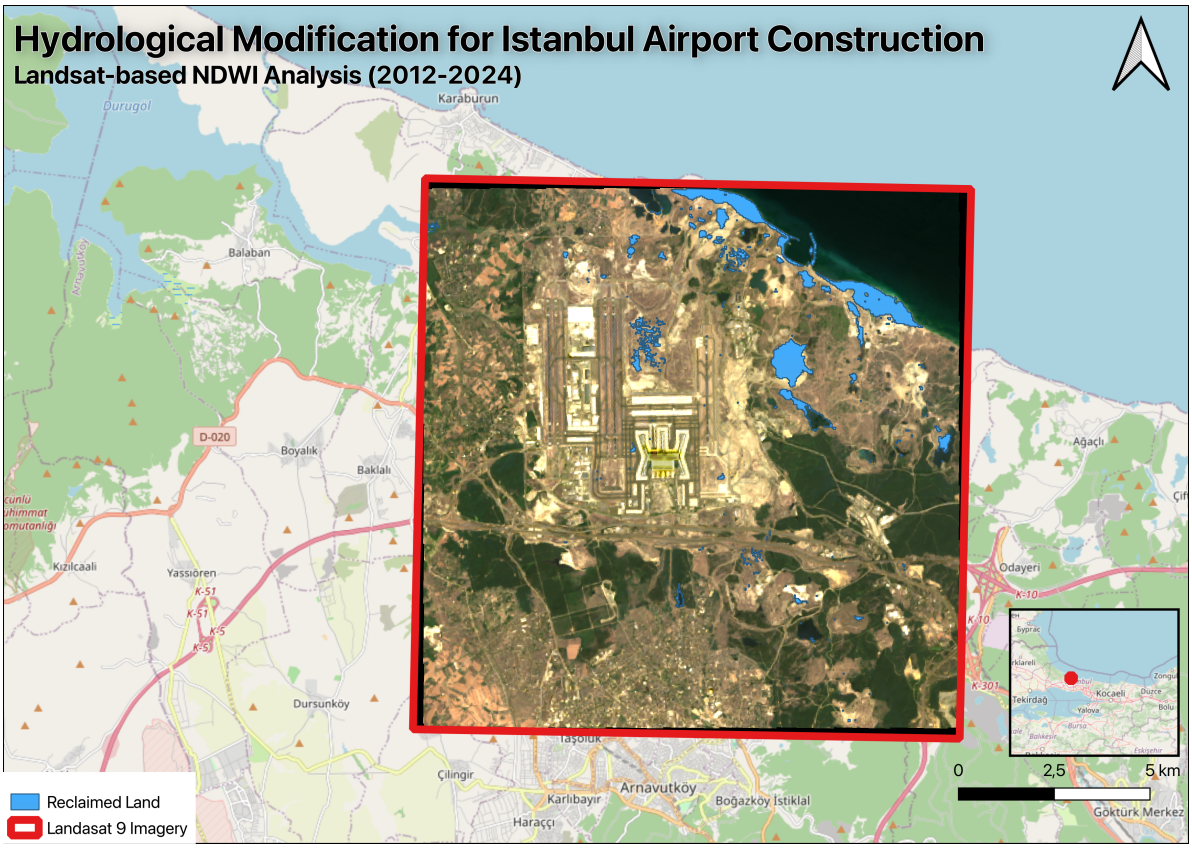


Figure 3: Hydrological modification for Istanbul Airport construction.

4 Conclusion

Using Landsat time-series analysis and NDWI-based water classification in Google Earth Engine, I quantified that **approximately 2.4 km^2 of water-covered areas were converted to terrestrial land** for Istanbul Airport construction between 2012 and 2024. This finding reflects substantial hydrological modification including both Black Sea coastal infilling and drainage of inland water bodies (lakes, ponds, wetlands). The analysis demonstrates the utility of remote sensing for monitoring large-scale land cover changes associated with infrastructure development.

References

[1] Istanbul Airport. *About Istanbul Airport*. https://www.istanbulairport.com/ (Accessed: January 2026).

[2] Wang, W., Liu, H., Li, Y., & Su, J. (2014). Development and management of land reclamation in China. *Ocean & Coastal Management*, 102, 415–425.

[3] McFeeters, S. K. (1996). The use of the Normalized Difference Water Index (NDWI) in the delineation of open water features. *International Journal of Remote Sensing*, 17(7), 1425–1432.

[4] Temmerman, S., Belliard, J.-P., & Slootmaekers, B. (2025/2026). *Geographical Information Systems (2001WETGIS)*. University of Antwerp, Faculty of Science, Department of Biology.

Google Earth Engine Project:
https://code.earthengine.google.com/e9c72566d45c96661dddc95ead646f67